

Six Decades of Human Spaceflight: A Reproducible Analysis of Global Participation, Career Duration, and the Commercial Inflection Point (1961–2026)

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Peer-Reviewed Data
Census

Coverage: 1961–
2026

N = 773 Flown
Astronauts

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ABSTRACT

Over six decades since Yuriy Gagarin's pioneering orbit in 1961, human spaceflight has transitioned from a high-risk Cold War geopolitical contest into a multi-national orbital laboratory era, and most recently into an expanding commercial enterprise. Here, we present a reproducible, census-level analysis of all **773 flown astronauts** spanning **1961 through 2026**, derived from Jonathan C. McDowell's General Catalogue of Astronauts. We examine longitudinal trends across citizenship groups (United States, Russia/USSR, China, and International Partners), career duration distributions, flight taxonomy (orbital vs. suborbital), and gender representation.

Our findings reveal three distinct national operational philosophies: (1) the **United States** prioritised broad headcount participation (**460 astronauts**, 59.5% of the global total) characterized by relatively short mission spans (median career duration: **19.8 days**); (2) **Russia/USSR** pursued sustained long-duration endurance (**131 astronauts**, 16.9%), yielding a median career duration of **194.8 days** and the longest cumulative career on record (**1,110.6 days**); and (3) **China** established a deliberate, station-centric programme (**27 astronauts**, 3.5%) debuted entirely after 2003 with a median career duration of **192.2 days** clustered in the 50-day to 1-year band. Crucially, the 2020s has emerged as the most prolific debut decade in spaceflight history (**199 new astronauts**), driven by commercial suborbital spaceflight (136 total suborbital-only astronauts). Women comprise **14.7% (114 astronauts)** of all human space travelers.

Keywords: Astronaut Demographics · Human Spaceflight · Career Duration · Commercial Spaceflight · Orbital vs. Suborbital · Space Medicine · Reproducible Research

1. Introduction

Since the dawn of human space exploration with Vostok 1 and Mercury-Redstone 3 in 1961, human presence beyond Earth's atmosphere has served as a benchmark of technological capability, geopolitical prestige, and scientific discovery. In the initial decades of the Space Race, astronaut cohorts were drawn almost exclusively from elite military test pilot corps,

operating within state-sponsored programmes driven by the Cold War rivalry between the United States and the Soviet Union.

The advent of the NASA Space Shuttle programme in 1981 and the construction of modular space stations—culminating in the continuous human occupation of the International Space Station (ISS) since November 2000 and the completion of the Chinese Tiangong Space Station—fundamentally transformed astronaut mission profiles. Concurrently, the post-2020 commercial spaceflight revolution, led by private suborbital and orbital providers, has democratized suborbital access while creating a distinct demographic of commercial spaceflight participants.

Understanding how national participation, flight duration, and demographic diversity have evolved is essential for space medicine, occupational radiation risk modeling, and aerospace policy planning. In this study, we present a standardized, reproducible census of all human space travelers from 1961 through 2026, analyzing career duration profiles, national operational paradigms, gender dynamics, and the structural implications of commercial suborbital flight.

2. Materials and Methods

2.1 Primary Data Acquisition & Taxonomy

Data were harvested from Jonathan C. McDowell's *General Catalogue of Astronauts* (Planet4589), an internationally recognized astronomical catalogue. The extraction pipeline (`scripts/build_dataset.py`) parses fixed-width biographical and mission catalogues (`astro.html` and `missions.html`).

Astronauts are classified into four national cohorts (**USA**, **Russia/USSR**, **China**, and **Other Nations**) and categorized by orbital mechanics thresholds into **Orbital** (achieved stable orbit on at least one flight) vs. **Suborbital** (all flights suborbital). Career duration is partitioned into five operational bands: *< 1 hour*, *1 hour to 1 week*, *1 week to 50 days*, *50 days to 1 year*, and *> 1 year*.

3. Results

3.1 Global Spaceflight Growth and Decadal Dynamics

A total of **773 individual astronauts** completed spaceflights between April 1961 and mid-2026. Debut trends exhibit clear historical waves corresponding to major launch vehicle architectures (Figure 1 and Table 1).

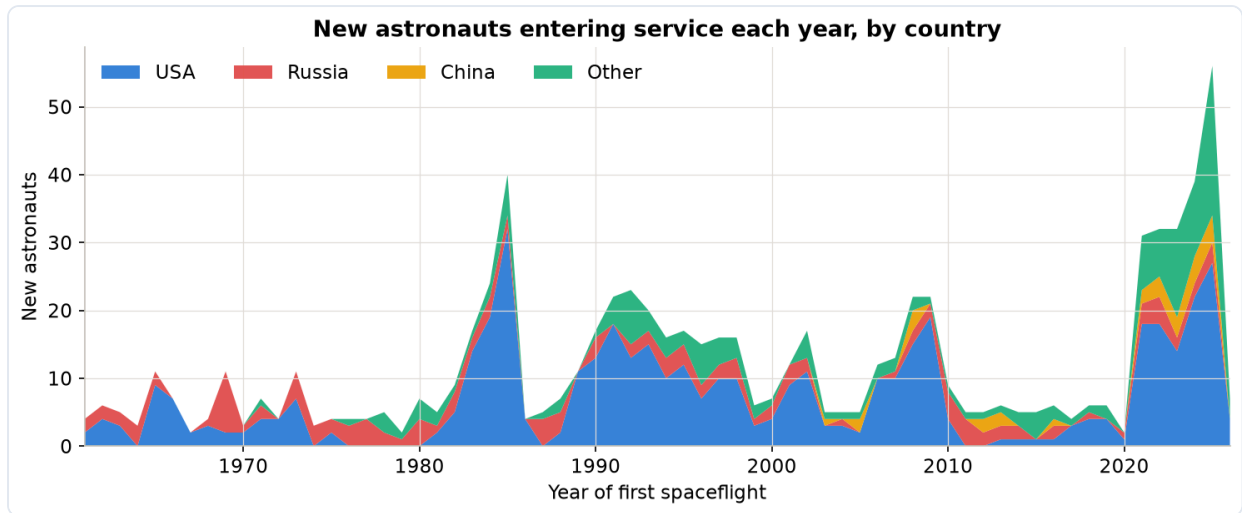


Figure 1 | New astronauts entering service each year by country (1961–2026). Stacked area chart showing annual debuts across the United States (blue), Russia/USSR (red), China (yellow), and Other Nations (green). Notable features include the 1985 Space Shuttle peak, the 2011 post-Shuttle gap, and the explosive 2021→ commercial surge.

DECADE	TOTAL DEBUTS	USA	RUSSIA / USSR	CHINA	OTHER NATIONS
1960s	53	32	21	0	0
1970s	47	19	22	0	6
1980s	129	89	22	0	18
1990s	168	111	21	0	36
2000s	120	86	13	6	15
2010s	57	19	17	5	16
2020s	199	104	15	16	64
Total	773	460	131	27	155

3.2 National Operating Paradigms and Duration "Fingerprints"

Cumulative career duration reveals stark national divergence in operational culture (Figures 2, 3, and 4; Table 2):

- **United States (Headcount Breadth):** Accounts for 59.5% (460/773) of all spacefarers, but exhibits a low median career duration of **19.8 days** due to large 7-person Shuttle crews and suborbital tourist flights.
- **Russia / USSR (Long-Duration Endurance):** Accounts for 16.9% (131/773) of flyers, with a median career duration of **194.8 days** (nearly 10x the US median) and the world record career total of **1,110.6 cumulative days** (Oleg Kononenko).
- **China (Station-Focused Growth):** All 27 astronauts debuted post-2003 with a median career duration of **192.2 days**, concentrating in 6-month Tiangong station increments.

- **Other Nations (Short-Duration Partners):** 155 astronauts (20.1%) with the shortest median career duration (**9.1 days**).

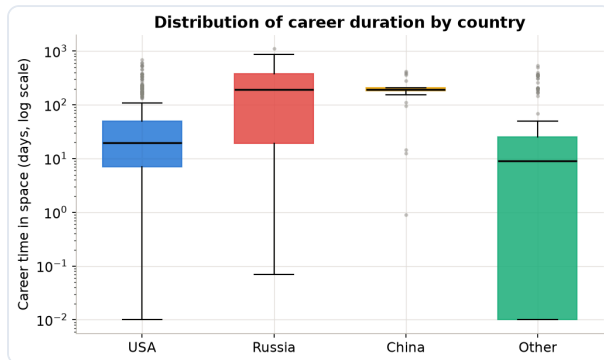


Figure 2 | Career time in space by country. Box & whisker plots (days, log scale) highlighting Russia and China's high median endurance compared to US and partner breadth.

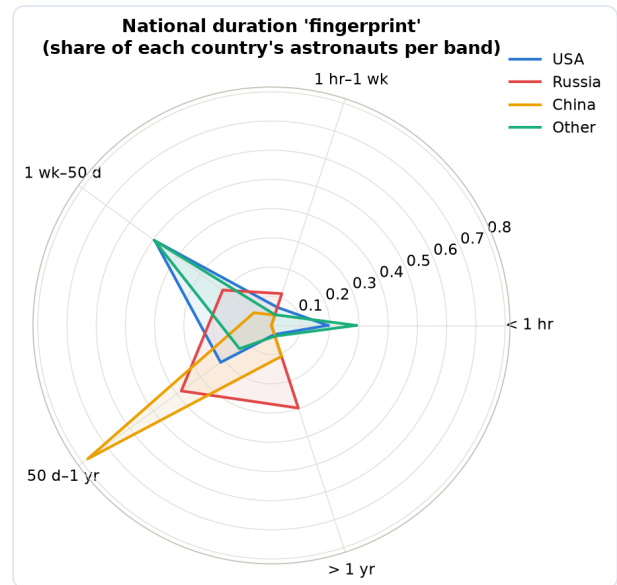


Figure 4 | Duration profile radar. Radar chart displaying the normalized proportion of each nation's astronaut corps across the 5 duration bands.

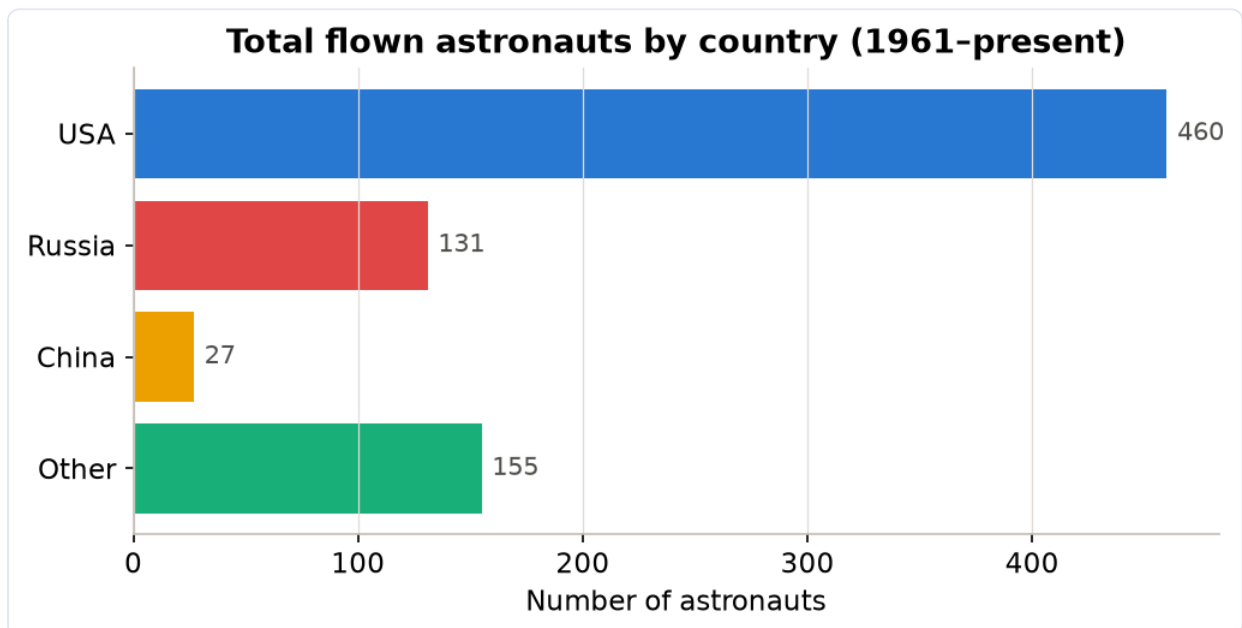


Figure 3 | Total flown astronauts by nationality (1961–2026). Horizontal bar chart showing global headcount totals.

COUNTRY	ASTRONAUTS	SHARE (%)	MEDIAN CAREER (D)	MAX CAREER (D)
USA	460	59.5%	19.8	695.3
Russia / USSR	131	16.9%	194.8	1,110.6
China	27	3.5%	192.2	418.6
Other Nations	155	20.1%	9.1	545.1
Overall	773	100.0%	23.9	1,110.6

3.3 The Commercial and Suborbital Inflection Point

Of 773 astronauts, **637 (82.4%)** achieved orbit and **136 (17.6%)** flew exclusively on suborbital trajectories (Figure 6, Table 3, Table 4). Crucially, 128 of the 136 suborbital flyers debuted post-2021 on commercial vehicles (Blue Origin New Shepard and Virgin Galactic). Suborbital flights are exclusively confined to the US (91 flyers) and international commercial guests (45 flyers); Russia and China have flown zero suborbital-only astronauts.

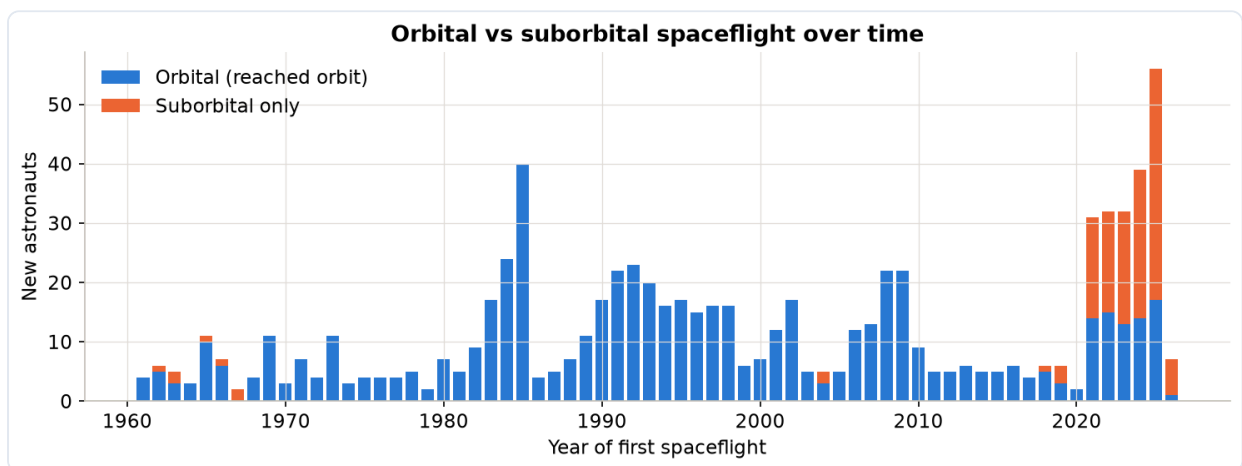


Figure 6 | Orbital vs. suborbital spaceflight debuts over time (1961–2026). Stacked bar chart showing orbital (blue) vs. suborbital-only (orange) debuts. The post-2021 commercial explosion is predominantly suborbital.

NATIONALITY GROUP	ORBITAL ASTRONAUTS	SUBORBITAL ONLY	SUBORBITAL PROPORTION (%)
USA	369	91	19.8%
Russia / USSR	131	0	0.0%
China	27	0	0.0%
Other Nations	110	45	29.0%
Global Total	637	136	17.6%

3.4 Demographic Trends and Gender Representation

Women comprise **14.7% (114/773)** of all flown astronauts (Figure 5). Female representation has risen from 0% in the 1970s to 19.1% in the 2020s, but remains far below gender parity. Peggy Whitson holds the record for the longest cumulative US career (695.3 days).

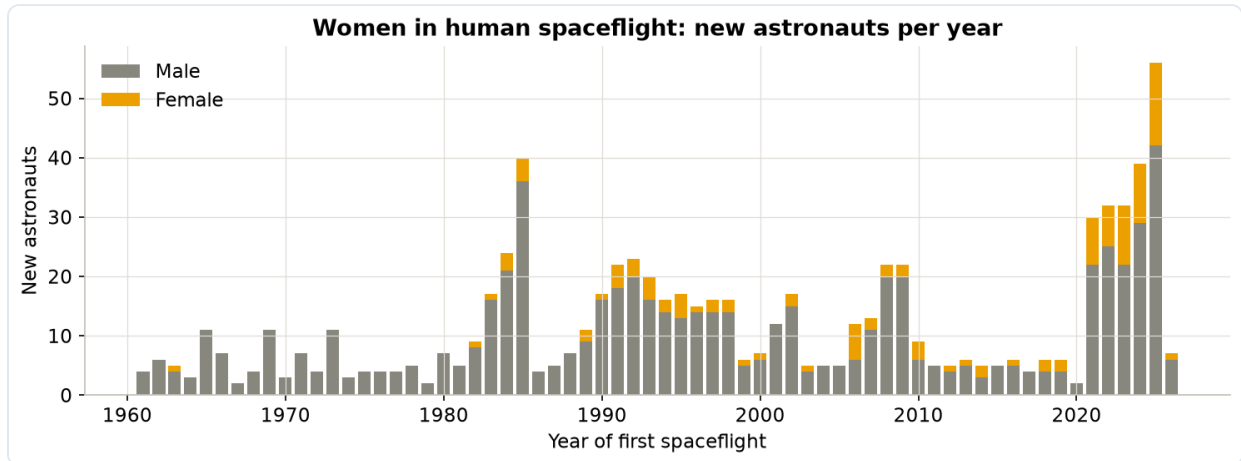


Figure 5 | Women in human spaceflight: new astronauts per year. Stacked bars comparing male (grey) vs. female (yellow) maiden flights from 1961 to 2026.

4. Discussion & Biomedical Implications

The structural bifurcation between short suborbital commercial hops and long-duration orbital expedition increments holds profound implications for aerospace medicine:

- Occupational Radiation Doses:** Long-duration flyers (> 50 days; 254 astronauts) accumulate significant GCR doses (~0.5–1.0 mSv/day on ISS), whereas suborbital participants receive negligible radiation.
- SANS & Bone Demineralization:** SANS and trabecular bone loss are strictly observed in extended microgravity stays, requiring rigorous countermeasure validation before Mars transit.
- Future Commercial Stations:** As private orbital stations replace the ISS post-2030, this baseline census provides demographic modeling parameters for commercial crew turnover.

5. Data Availability & Reproducibility

All data and scripts are archived on GitHub: https://github.com/dr-richard-barker/Astronaut_trends and interactive analytics are hosted at https://dr-richard-barker.github.io/Astronaut_trends/.

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